

Technical Deliverable: DoorList

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December 9, 2025

1 Introduction

Coming to the University of Virginia, we didn't think an unfamiliar event app would end up being such a significant part of our college life, but DoorList quickly changed that. It didn't take long to see how much people rely on it: from academic and club meetings to Greek life events, rush activities, bar tabs, fundraisers, and even just casual hangouts. Whether you're hosting or attending, DoorList makes event organization much easier.

DoorList is used by college students, organizations, and event hosts to manage and check people into events easily. Traditionally, organizations or event hosts have relied on physical check-in lists or wrist bands. DoorList replaces this traditional approach with a digital QR code used for entry. It is deeply rooted in UVA culture, having been created by a UVA alumnus who recognized the need to streamline attendance tracking for both hosts and attendees. Since its launch at UVA, the app has expanded to other universities such as the University of North Carolina and the University of Texas.

With DoorList being such an integral part of UVA's campus culture, we saw value in analyzing its impact, especially since it is a tool we personally rely on in our own organizations. We believe our analysis can help organizations better understand their hosting patterns and improve how they plan events and manage resources, while also providing DoorList with insights to enhance its platform. Through our findings and suggestions, we aim to benefit not only UVA students but also those at other universities as DoorList continues to grow.

To jumpstart our project, we reached out to some fraternities to request access to their data. We soon realized there would be numerous obstacles to obtaining it, and any data we did receive would likely be incomplete. In addition, our sample size would be quite small and inaccurate, for it would only include some fraternities. This approach would not give us access to individual hosts' data or other types of hosts, like academic clubs or sororities.

Once we realized that acquiring data from fraternities was not a viable option, we scheduled a call with DoorList. We connected with their team and arranged a Zoom meeting with UVA alumnus Mitchell Whalen, a growth associate. After discussing our capstone goals, we were granted access to DoorList's sensitive data upon signing an NDA to ensure the privacy of attendees, hosts, and organizations. Accessing the data directly from DoorList was ideal, as it ensured accuracy and reliability. We met with Mitchell multiple times to review the dataset, gaining a clear understanding of each variable and how to use the information effectively.

2 Data

The data we received is at the event level, meaning each row represents a single event rather than an individual attendee, which allowed us to focus on how events differ from one another. It covers a wide range of UVA DoorList events, including big parties, meetings, date functions, small gatherings, and other social events, spanning from August 5th, 2024 to September 8th, 2025. Although the data was in JSON format, we decided to use the information available and turn it into something workable for our analysis.

The data contained 1,213 rows of UVA events. There were 117 columns in the original dataset, including many predictor variables and our response variable, `thruDoor`, which lists the IDs of people who attended an event via DoorList’s scanning feature. After discussions with DoorList representatives, we concluded that including all 116 predictor variables in our model would not be practical. Many predictors were irrelevant to our analysis, some contained mostly missing values, others were obsolete and no longer in use, and, most importantly, having too many predictors would hinder our model’s performance. Therefore, we decided to filter the dataset to only include 13 of those original 116 predictors. Additionally, DoorList representatives gave us a list of organization names that were associated with their development team’s test cases. We filtered the dataset by removing these test cases and any cancelled events.

Next, in order to better format the JSON data for our analysis, we had to parse through fields and make additional columns with these extracted values. First, we parsed through the `thruDoor` column, our response variable, by counting the number of commas separating the different attendee IDs and adding one to accurately represent the total number of attendees. If there was an empty list of IDs, the attendee count for that event was set to zero. After parsing `thruDoor`, we created a new column, `attendee_count`, which counts the number of attendees at each event. We followed the same approach for other columns like `approvedID`, which counts the number of approved IDs for an event and totals them in a column named `numApprovedID`. For other columns that we needed to parse through, such as `requestedPlusOnes`, `going`, `maybe`, `notGoing`, `allowedOrgs`, and `orgPlusOnes`, we followed similar procedures to count the IDs and made columns such as `totRequestedPlusOnes`, `RSVPyes`, `RSVPmaybe`, `RSVPnotGoing`, `numAllowedOrgs`, and `orgPlusOnes_num` respectively. Finally, we changed the time variable from UTC format to Eastern time and extracted both the hour and day of the week from each timestamp to make the columns `event_hour` and `event_day`. `event_hour` is represented by a number from 0 to 23, and `event_day` is

represented by the day of the week. With these new variables, we were able to make our time-based analyses more interpretable and uncover patterns in event activity throughout the day and week.

During our initial exploratory data analysis, we observed that many events in the dataset had attendee counts of zero. To address this, we decided to focus on a subset of events: those with at least one invited organization, regardless of attendance size, and those without invited organizations only if they had ten or more attendees. We made this choice because many events appeared to be “joke” events created by users that never actually took place. However, we still included events with zero attendees if at least one organization had been invited, since some legitimate events are listed on DoorList but have no scans, a pattern we have observed from experience. This approach allowed us to retain valid but unscanned events while filtering out those that were likely invalid. After filtering, there were 635 events remaining in our dataset.

Based on our understanding of UVA events and DoorList, we decided to add a new variable, `event_type`, to classify each event. Since our filtered dataset included 635 events and our team consists of four members, we divided the events among ourselves and individually classified each type. We developed eight categories to classify each event: big party, mixer (get-together with 2-4 frats/sororities), me + 3 (fraternity/sorority party where each person in the organization receives three guest invites), rush event, meeting, date function, small event, and an “other” category to capture any remaining events. We relied on our judgement and experience as UVA students to assign each event to the most appropriate category. To classify events in our model, we created a data dictionary that maps specific keywords from event names to their corresponding categories. For example, some key words for date functions are “dfunc”, “date function”, “df”, etc.

Since our data is structured at the event level, the main research question we chose to study is: *How many people actually attend an organization’s events?* We decided to research this question because it closely aligns with DoorList’s purpose. The app is designed to track attendance and aid hosts in managing who attends their events. Rather than measuring how “successful” an event is, our goal is to predict how many people will attend so hosts can plan more effectively. By examining factors such as event type, time and day, number of invited organizations, RSVP responses, among others, we can hopefully identify characteristics closely associated with event turnout. Understanding these patterns can help organizations plan their events more effectively and make smarter decisions about scheduling, capacity, and preparation.

While the dataset we received from DoorList was substantial, there were still a few components we wish we could have included. The main limitation was that our data was at the event-level instead of the individual-level. Having individual-level data would have allowed us to explore student engagement on Grounds in a more specific way. For example, we could have potentially found patterns like whether friends or people in the same organizations tend to go to similar events, if students in certain groups prefer particular event types over others, and how people's attendance habits change over time. It would have been interesting to study and include how individual and group dynamics contribute to event attendance.

We also wish we had access to more than one year of data. Multiple years of data would have provided more information about how events and attendance have changed over time and whether any long-term patterns exist. Furthermore, it would have been valuable to look at recurring events to observe if attendance stayed consistent or shifted from year to year, or if certain event types became more or less common over time. With additional years of data, we also could have examined whether hosting activity varies by semester and whether certain parts of the year consistently see higher event activity and attendance than others.

In addition, there were a few other pieces of information that would have been beneficial for our analysis. For example, weather data may have helped explain variation in turnout, since factors like rain, cold temperatures, or warm, sunny days can affect whether people choose to attend an event. It also would have been useful to know if there were other non-DoorList events occurring at the same time, since those could draw attendance away from DoorList events. Finally, how events were promoted, such as whether they were advertised more publicly across Grounds or mainly within smaller groups or organizations, would have provided valuable insight. This additional information could have clarified what factors drive attendance and how external conditions might influence attendance patterns.

After cleaning and preparing the data, we explored some general patterns in attendance and related variables to better grasp the information before entering the modeling phase.

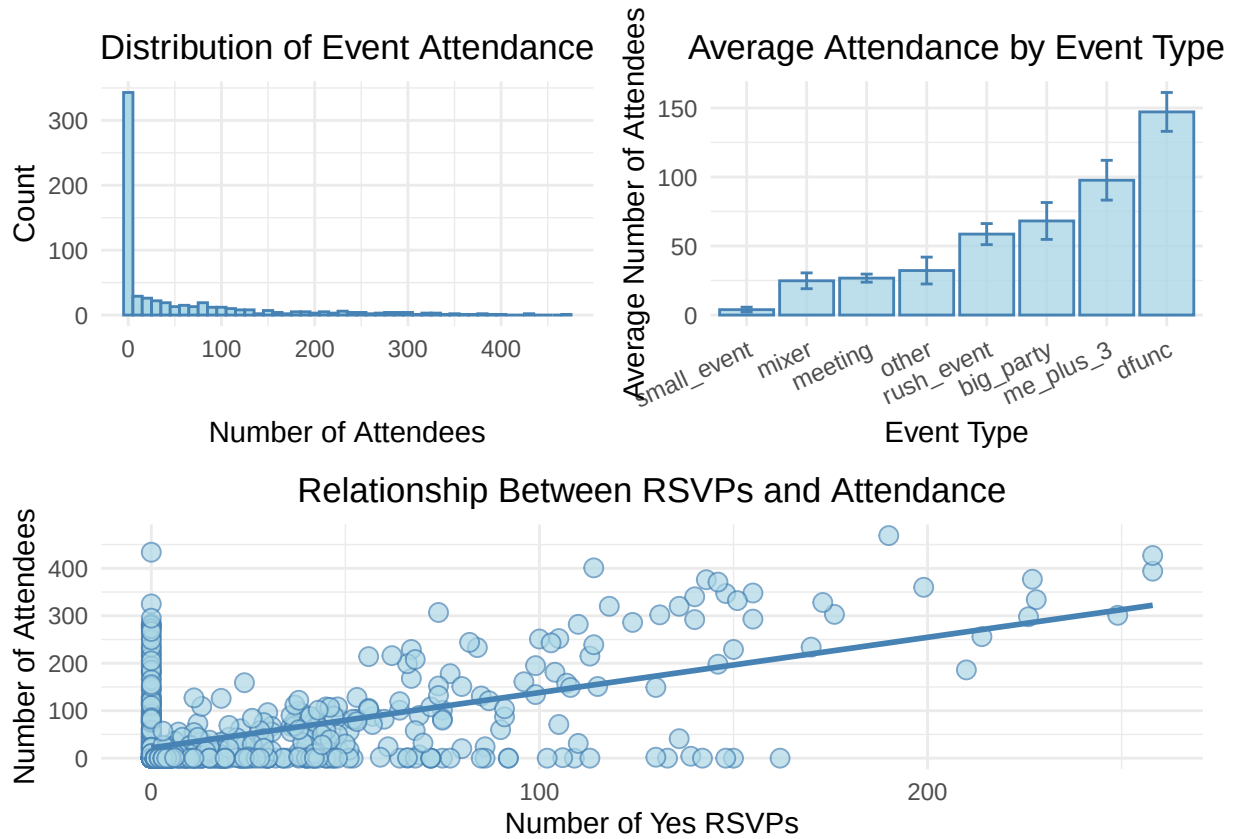


Figure 1: Attendance Exploratory Plots

Demonstrated in the top-left plot in Figure 1, the distribution of attendance was strongly right-skewed, with most events drawing only a handful of people and only a small subset of events with high turnout. Attendance also varies by event type, with small events, mixers, and meetings averaging fewer attendees, while date functions and me-plus-three events attract larger crowds, which is shown in the “Average Attendance by Event Type” bar chart.

The scatterplot between RSVPs and attendance shows a clear positive relationship, in which events with more “Yes” RSVPs tend to have higher attendance. The cluster of points along the x-axis at zero likely represent events where organizations did not scan attendees, resulting in an attendance of zero despite having RSVPs. Conversely, the points along the y-axis at zero likely reflect events with no RSVPs where attendees still showed up to the event.

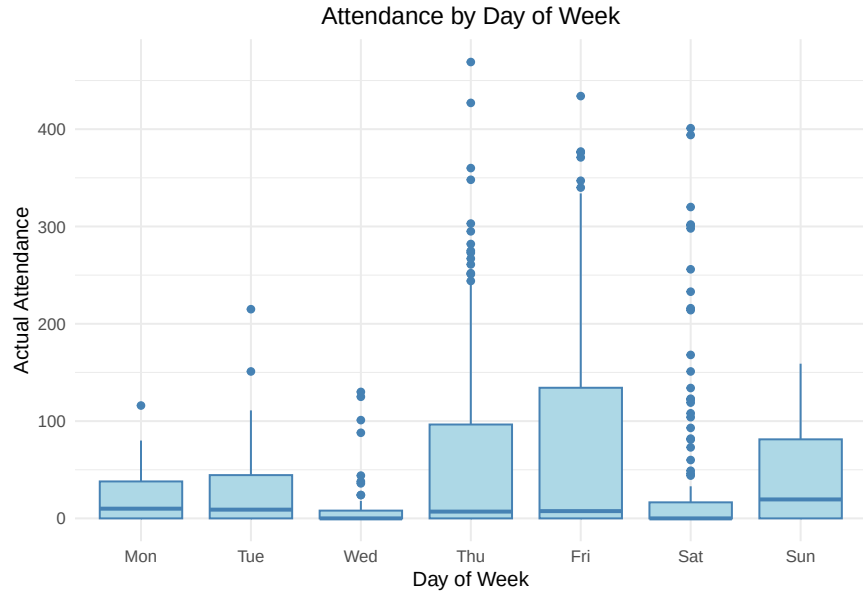


Figure 2: Attendance by Day of Week

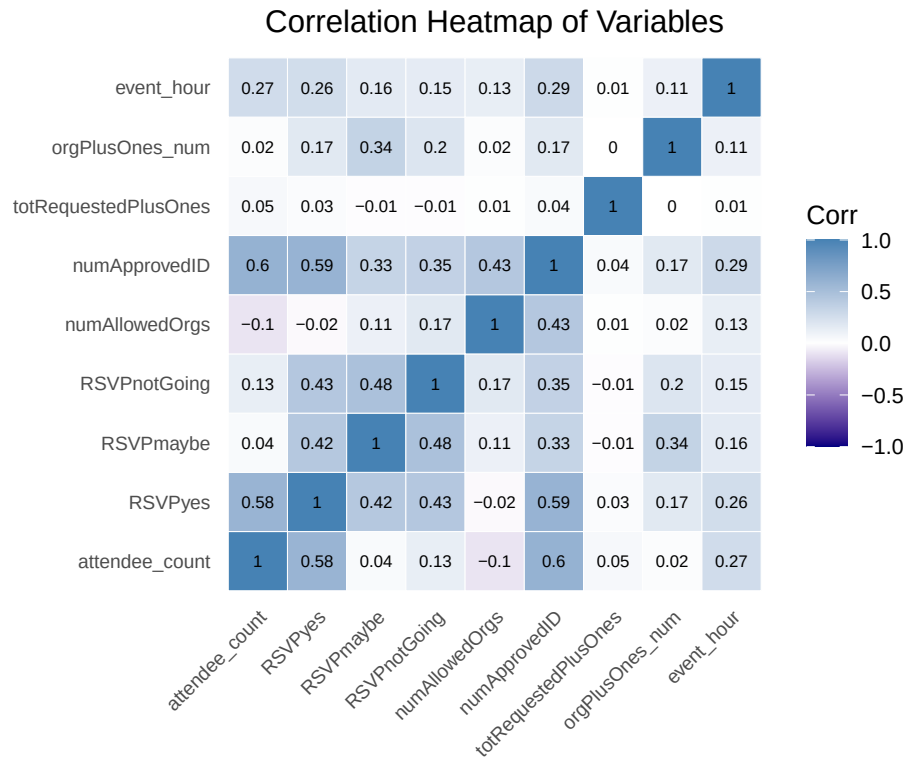


Figure 3: Correlation Heatmap

Figure 2 displays the distribution of attendance across days of the week. Thursday and Friday show the greatest variability in attendance, with the largest IQRs, while Saturday has a smaller IQR but several high outliers. Sunday also shows moderately wide variation. Based on these patterns and the context of UVA event culture, we grouped days into two categories: weekdays and weekends, defining weekends as Thursday at 12 AM through Sunday at 2 AM, with all remaining days classified as weekdays. This is due to the fact that although UVA classes are in session Monday through Friday, there is an overall consensus that Thursday indicates the start of the weekend, and this was proven in the distribution above.

Finally, Figure 3 presents a correlation heatmap showing which numeric variables are most related to attendance. The strongest positive correlations were with `numApprovedID` (0.6) and `RSVPyes` (0.58), indicating that events with more approved IDs and more “Yes” RSVPs tend to have higher turnout. Attendance was also weakly correlated with `event_hour` (0.27), suggesting slightly higher turnout for later events. All other correlations were minimal, implying that they have little linear relationship with attendance. These insights helped guide which variables to prioritize in our modeling.

3 Analysis

Regarding our analytical approach, we utilized multiple linear regression with stepwise model selection based on the Akaike Information Criterion (AIC) in R. This method provided us a systematic way to identify the model that best explains variation in the response variable while maintaining a good balance between fit and complexity. Specifically, our analysis began by fitting a full model that included all theoretically relevant predictors and possible interaction terms. The `stepAIC()` function then iteratively added and removed predictors, calculating the AIC at each step and selecting the model with the lowest AIC value. Including the interaction term between `numAllowedOrgs` and `numApprovedID` allowed the model to capture how the combined effect of the number of organizations invited and the total number of people invited influences attendance, which may improve model accuracy.

Multiple linear regression was selected because it offers a clear framework for quantifying relationships between multiple explanatory variables and a continuous dependent variable. It allows for direct interpretation of coefficients, which aligns with our goal of making the results intuitive and easy to communicate with our audience. It also clarifies which predictors contribute most strongly to the outcome and in what direction. The stepwise AIC approach is

particularly appropriate here because it balances model complexity with explanatory power, automatically penalizing models that include unnecessary predictors. This method ensures that our final model is both statistically robust and interpretable.

Before finalizing our model by using multiple linear regression and stepwise selection, we also considered ridge regression. Given our limited number of observations, we thought this would be a logical method to improve our prediction model. Ridge regression optimizes the model by shrinking the beta coefficients toward zero. We decided to choose the regularization parameter automatically using cross-validation with the `cv.glmnet()` function. Although we expected ridge regression to enhance our model, its R-squared value was lower than that of our multiple linear regression model. We also preferred multiple linear regression for its greater interpretability, since ridge regression shrinks coefficients. Additionally, ridge regression is most beneficial when there is high multicollinearity between variables, which was not a concern in our dataset.

Although multiple linear regression has many strengths, it also has some notable limitations. The method assumes a linear relationship between predictors and the dependent variable, meaning it may fail to capture nonlinear trends or threshold effects. It is also highly sensitive to outliers and influential data points, which can distort model estimates and lead to misleading conclusions. Furthermore, stepwise selection can sometimes lead to overfitting, particularly when many predictors are tested, and underfitting when potentially meaningful variables are excluded. In our data specifically, some variables show skewed distributions or restricted ranges, which could affect coefficient stability. In addition, verifying that our model satisfied the assumptions of multiple linear regression was important, for unmet assumptions could lead to greater limitations in the analysis.

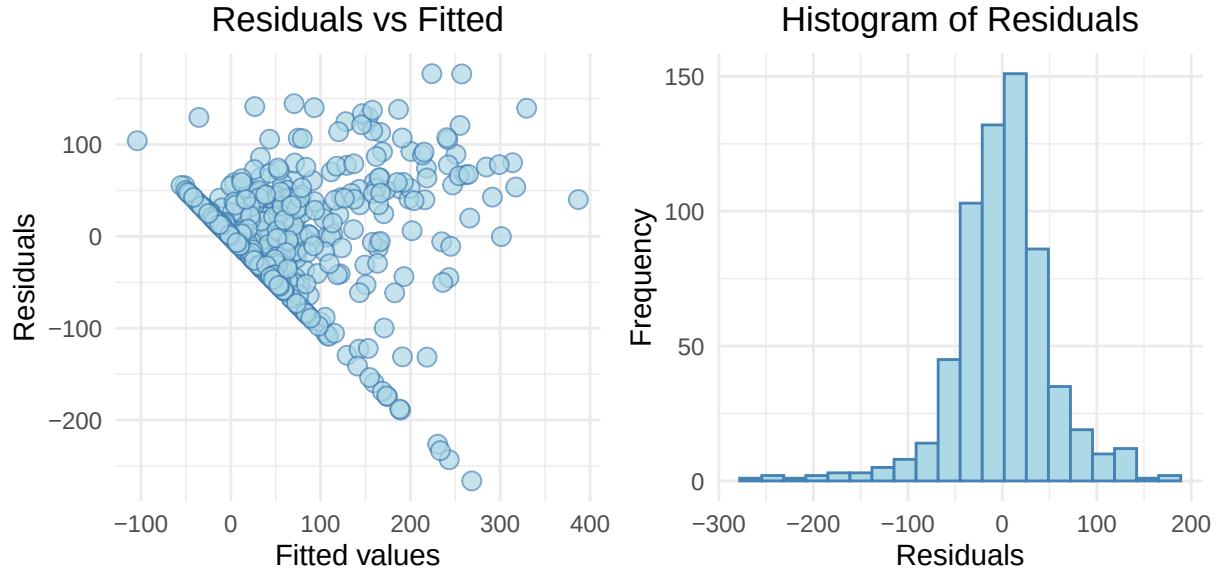


Figure 4: Assumption Plots

Regarding the linearity assumption, we assessed this concern utilizing a fitted-versus-residual scatterplot. Demonstrated above in the left plot of Figure 4, the residuals create a sharp, downward-sloping edge. This occurs because actual attendance cannot go below zero, while the fitted values from the model can. Therefore, despite the sharp boundary, the rest of the points do not follow any systematic pattern, suggesting that the relationships in the model are generally linear. Independence of errors was examined using the Durbin-Watson test, which resulted in an insignificant p-value of 0.2539. This indicated that residuals were not significantly autocorrelated, thus supporting the assumption of independence. Regarding the assumption of constant error variance, residual plots revealed strong heteroscedasticity, suggesting that the spread of residuals increases with fitted values. This may indicate model misspecification or unaccounted variability, meaning that the assumption of constant error variance was not met. For example, if the variance of residuals increases with the number of people invited, the model is less certain about predictions for high values. Next, the normality of residuals assumption was satisfied, as shown by the histogram on the right in Figure 4, which displays an approximately normal distribution. Lastly, multicollinearity diagnostics revealed that all of our predictor variables had low VIF values, confirming that this assumption was met as well.

4 Interpretations & Recommendations

Our final model is:

$$\begin{aligned} \text{attendee_count} = & -14.03 + 0.454 \cdot \text{RSVPyes} - 3.817 \cdot \text{RSVPmaybe} - 4.271 \cdot \text{RSVPnotGoing} \\ & - 17.757 \cdot \text{numAllowedOrgs} + 0.500 \cdot \text{numApprovedID} + 1.412 \cdot \text{event_hour} \\ & - 7.971 \cdot \text{day_typeweekend} + 27.392 \cdot \text{event_typedfunc} \\ & - 39.275 \cdot \text{event_typeme_plus_3} - 18.767 \cdot \text{event_typemeeting} \\ & - 25.989 \cdot \text{event_typemixer} + 6.830 \cdot \text{event_typeother} \\ & + 9.943 \cdot \text{event_typerush_event} - 1.138 \cdot \text{event_typesmall_event} \\ & - 0.078 \cdot (\text{numAllowedOrgs} \times \text{numApprovedID}) + \varepsilon \end{aligned}$$

Based on our model results, several variables appear to have strong and statistically significant relationships with event attendance. RSVP behavior is one of the most influential predictors, with each additional “Yes” RSVP associated with an estimated increase of about 0.45 attendees and each “Maybe” RSVP associated with roughly 3.82 fewer attendees on average, likely reflecting overestimation of attendance among preliminary respondents. In practice, this suggests that hosts should treat “Yes” RSVPs as meaningful indicators of turnout, while “Maybe” RSVPs are negatively associated with turnout and may indicate that fewer attendees will ultimately arrive.

The number of approved individual invitations (**numApprovedID**) shows a strong positive effect, with an estimated coefficient of 0.50. This suggests that, on average, about half of those who are invited to a given event actually attend. However, the interaction between the number of organizations invited (**numAllowedOrgs**) and the number of individuals invited is negative and significant, suggesting diminishing returns: as more organizations are invited, each additional approved invitation contributes less to overall attendance. This suggests that inviting additional organizations does not necessarily increase turnout in proportion to the number of people invited. Since the approved ID list already includes members of all invited organizations, adding more organizations may broaden the invitation pool without meaningfully increasing actual attendance. In practice, hosts may experience reduced engagement per invited person as the number of invited organizations grows.

Timing also has a moderate but meaningful impact, such that events held later in the day (**event_hour**) tend to attract slightly higher attendance, consistent with expectations for

many of the social activities in our dataset. Weekend events show somewhat lower average attendance than weekday events; however, this difference is not statistically significant, suggesting that event timing may be less influential than other characteristics.

Regarding event type, certain categories (e.g., “dFunc” events) show significantly higher attendance, while others such as mixers or meetings draw smaller crowds. These differences highlight the value of understanding how event purpose influences turnout and can inform targeted planning for different event styles. Event planners can leverage these patterns by anticipating higher turnout for certain event types and managing resources accordingly.

Overall, our regression results reveal several recommended strategies for increasing attendance across events. First, maximizing positive RSVP engagement is critical: increasing the number of “Yes” RSVPs substantially raises expected attendance, while “Maybe” and “Not Going” RSVPs have strong negative impacts. Organizations should therefore focus on clearer communication and encourage users to RSVP “Yes” rather than defaulting to “Maybe.” Additionally, more exclusive events regarding the organizations invited improves expected turnout, as a smaller `numAllowedOrgs` not only has a positive direct effect but also removes the negative interaction penalty between `numAllowedOrgs` and `numApprovedID`. Once this is minimized, an increased `numApprovedID`, for example, inviting less organizations with a greater number of members as opposed to more organizations with less members, further boosts attendance. Events held later in the day are also expected to draw larger crowds, with predicted attendance increasing the later an event is hosted, meaning events held closer to midnight experience a higher attendance. Finally, our model indicates that date functions produce the highest positive effect among event types, suggesting that structuring events more similarly to date functions may enhance turnout.

While these recommendations follow directly from the model’s coefficients, it is important to acknowledge several contextual limitations when interpreting them. For instance, the negative coefficient on weekend events may reflect the types of events held during the week rather than the true effect of day type itself. Many weekday events in our dataset are academic and may have required attendance. We can also make the assumption that events such as these also tend to enforce scanning more consistently, making their attendance counts more complete and reliable. In contrast, weekend events are typically optional social activities, where scanning may be inconsistently enforced or not emphasized at all depending on the event, leading to undercounted attendance and greater variability. Recommendations such as minimizing the number of allowed organizations or favoring certain event types must also be viewed realistically: some events require multiple organizations, and not all events can,

or should, be reframed as date functions. Additionally, event variety itself is a driver of engagement; if date functions became overly common, their attractiveness could likely diminish. Finally, because attendance in our dataset reflects scanned guests rather than true participation, some patterns may capture scanning behavior rather than actual turnout. These considerations emphasize that while the statistical model offers valuable direction, applying its insights requires balancing logistics, event priorities, and the overall context of UVA culture.

5 Results

To evaluate how well the model predicts attendance, we first summarized overall error metrics. According to the mean absolute error (MAE), our model is off by about 37 people on average, which is influenced by larger errors in the dataset. The median absolute error is about 26, showing that for a more typical event, ignoring those extreme outliers, the prediction is off by roughly 26 people. The root mean squared error (RMSE) is 52.9, which is much larger than the MAE of 36.9. Since the RMSE is notably higher, this suggests that there are some events with considerably large prediction errors, which inflates RMSE. This is expected for attendance data, since the number of attendees can vary widely across events. Since many events have zero attendees or very low attendance likely due to hosts not scanning, we chose not to use relative error. Relative error compares the size of the prediction error to the actual attendance. When actual attendance is zero or extremely low, this number can become small or undefined, which makes these results unreliable. Because of this, the absolute error metrics provide a more interpretable way to evaluate the model's performance.

The adjusted R^2 is approximately 0.64, meaning that the model explains about 64% of the variation in event attendance after accounting for the number of predictors. This shows that the model is picking up real patterns in the data, but there is still some variability that is not explained by noise or other factors not included in the model.

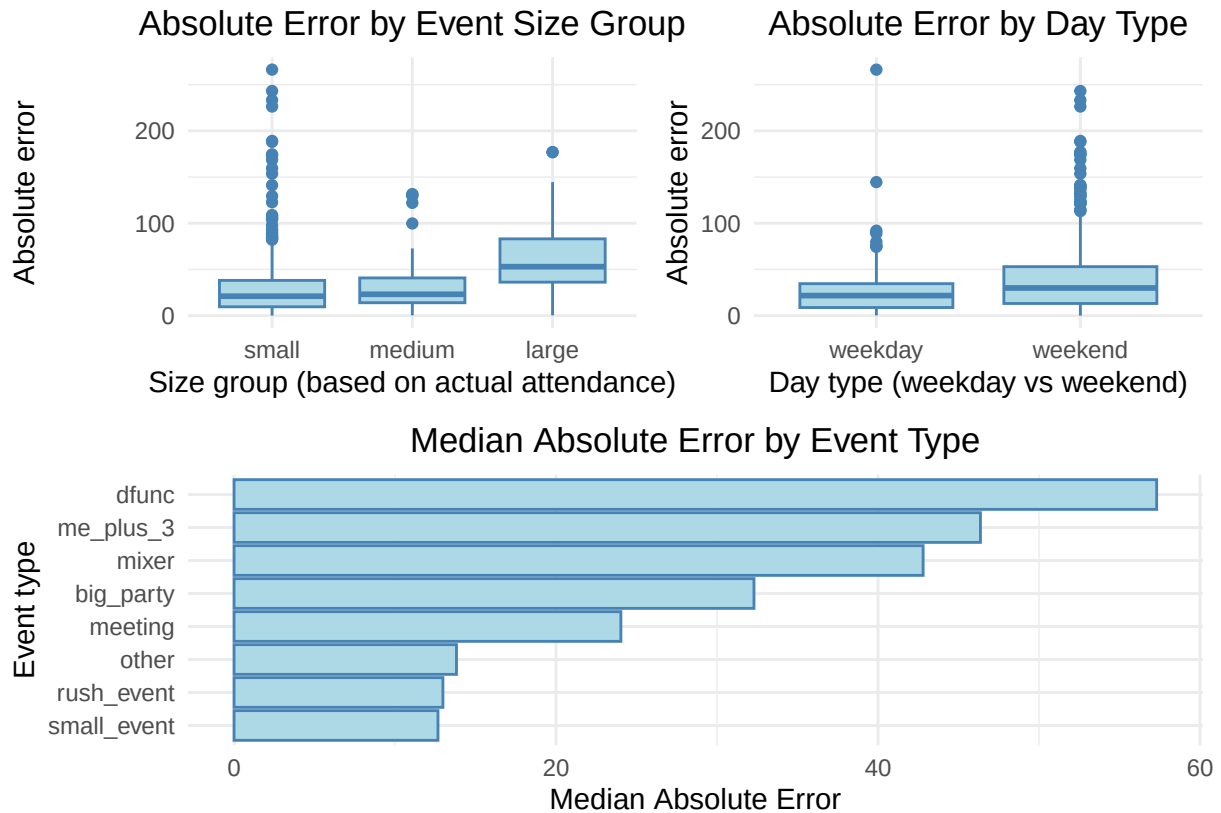


Figure 5: Error Distribution Plots

We compared model performance across event sizes, and there were noticeable differences in how accurately the model predicted attendance. Small events (0 to 30 attendees) had the lowest errors, with a median absolute error of about 21 attendees, and medium events (31 to 100 attendees) were predicted with similar accuracy, with a median absolute error of about 23 attendees. As shown in the boxplot, both small and medium events have fairly tight spreads, meaning their prediction errors fall in a small range and the model is usually off by a similar amount each time. Large events (100+ attendees) are considerably harder to predict. The median absolute error increases to about 53 and the boxplot shows a wider spread. This indicates that the model is less reliable for larger events because attendance numbers can vary more and there is greater uncertainty, leading to larger errors.

Similar to event size, when comparing attendee prediction accuracy across event types, there are clear differences in how well the model performs. Smaller events are the easiest to predict. For example, small events (small get-togethers) and rush events have the lowest median absolute errors of approximately 13 attendees, indicating that these events attract a similar

number of people each time. Therefore, the model has an easier time predicting them. Meetings are also fairly more predictable, with a median error of about 24. In contrast, the model has a more difficult time predicting the larger, more social events. Big parties, mixers, me+3's, and date functions all have higher errors, with median absolute errors of approximately 32, 43, 46, and 57, respectively. This means their attendance varies more, and the model has a harder time predicting them accurately. Overall, the model is more accurate for event types with consistent attendance and less accurate for those where turnout tends to fluctuate more.

To further understand how the model performs across different prediction levels, not just actual attendee counts, we also grouped events by their predicted attendance. Low prediction events, or events with a predicted attendance under 30, have a median absolute error of about 13, which suggests fairly consistent and accurate predictions. Medium prediction events, or events with a predicted attendance from 31 to 100, have a median absolute error of around 37. Lastly, for high prediction events, or events with predicted attendance over 100, their median absolute errors increase to approximately 66. This pattern of increasing errors shows that the model's accuracy decreases when its predicted attendance gets larger.

In addition, the model performs differently for weekday and weekend events. Weekday events have a lower median absolute error of around 22, whereas weekend events have a higher median absolute error of approximately 30. The boxplot also reflects this pattern, with weekend events having a wider spread of errors. This comparison suggests the model is better at predicting weekday events than weekend events. This aligns with the `event_type` results, in which many of the larger and more variable event categories occur on the weekends.

We then checked whether the final model tends to overestimate or underestimate attendance. The results showed a relatively even split, with about 48% of predictions higher than the actual attendance and about 52% lower. This suggests the model does not consistently predict too high or too low, meaning errors are fairly balanced.

6 Future Work

Several opportunities exist to extend and strengthen this analysis. First, further development of our event type classification dictionary would help standardization across organizations and new data. Given that our current event type dictionary relies heavily on UVA-specific context, future work should include both refining these labels for UVA culture and exploring how similar, context-appropriate classification systems could be developed for other DoorList-using universities. Establishing a consistent and scalable framework would improve accuracy and long-term usability of the variable.

Another aspect of future work involves revisiting a secondary question we were unable to pursue: understanding why some organizations adopt DoorList and use scanning features consistently while others do not. Addressing this question could help DoorList tailor its product design to encourage more reliable usage of scanning features. Our initial exploration revealed that answering this question currently requires assumptions about user behavior that are not reliably measurable in the dataset. Future work could focus on gathering behavioral data, such as scanning compliance logs or surveyed motivations, to reduce reliance on assumptions and allow for a more data-driven analysis. Furthermore, this may allow us to more accurately predict true attendance patterns rather than merely scanning patterns.

Finally, obtaining additional years of UVA DoorList data would enable us to evaluate model stability over time, test the generalizability of our findings, and treat our current model as a training phase. With multiple years of data, we could validate and refine the model, detect meaningful trends, and ensure that recommendations remain accurate.

7 Appendices

7.1 Appendix I: Data Dictionary

variable	definition
thruDoor	Array of all User IDs who checked-in. Could be via scan, location-based check-in, or self check-in.
name	Name of event.
allowedOrgs	OrgIDs that were invited to event.
eventCancelled	If event was deleted.
orgFBid	OrgID of primary hosting group. Could be both a group or just a user.
time	Start time of event.
orgName	Name of primary hosting group. Name associated with orgFBid.
approvedID	Array of all UserIDs on the guest list.
requestedPlusOnes	UserID and how many +1s they requested.
visibility	If event is Public or Private.
orgPlusOnes	Group ID and how many +1s each member gets.
going	Users who RSVPd going.
maybe	Users who RSVPd maybe.
notGoing	Users who RSVPd not going.
attendee_count	Count of thruDoor. Total number of users who checked in to event.
numApprovedID	Count of allowedOrgs. Total number of users invited to event.
totRequestedPlusOnes	Count of requestedPlusOnes.
RSVPyes	Count of going. Total number of users who RSVPd going.
RSVPmaybe	Count of maybe. Total number of users who RSVPd maybe.
RSVPnotGoing	Count of notGoing. Total number of users who RSVPd not going.

numAllowedOrgs	Count of allowedOrgs. Total number of organizations that were invited to event.
orgPlusOnes_num	Total number of +1s each member of hosting organization gets.
event_hour	Start hour of event on 24-hour clock.
event_day	Day of the week (Sun–Sat) event is being hosted.
event_type	Type of event (small_event, big_party, meeting, mixer, me_plus_3, other, rush_event, dfunc).
day_type	Weekend (Thursday through Sunday at 2am) or Weekday (Sunday 2:01 am through Wednesday at 11:59pm).

7.2 Appendix II: Code

Due to the terms of our NDA with DoorList, we are unable to include data-cleaning code or any code that interacts with sensitive or identifying information. Only the final modeling code is provided in this appendix, as it does not expose protected data.

```
fullmodel <- lm(attendee_count ~ RSVPyes + RSVPmaybe + RSVPnotGoing +
  numAllowedOrgs + numApprovedID + orgPlusOnes_num +
  event_hour + day_type + event_type +
  numAllowedOrgs * numApprovedID,
  data = filtered_uva_doorlist)

step_model <- stepAIC(fullmodel, direction = "both", trace = TRUE)
```

```
## Start:  AIC=5072.05
```

```
## attendee_count ~ RSVPyes + RSVPmaybe + RSVPnotGoing + numAllowedOrgs +
##   numApprovedID + orgPlusOnes_num + event_hour + day_type +
##   event_type + numAllowedOrgs * numApprovedID
```

```
##
```

	Df	Sum of Sq	RSS	AIC
## - orgPlusOnes_num	1	3750	1775643	5071.4
## <none>			1771893	5072.0
## - day_type	1	5835	1777728	5072.1
## - RSVPnotGoing	1	7815	1779708	5072.8
## - event_hour	1	16172	1788065	5075.8
## - numAllowedOrgs:numApprovedID	1	55789	1827682	5089.7
## - RSVPyes	1	116879	1888772	5110.6
## - event_type	7	194444	1966337	5124.2
## - RSVPmaybe	1	178440	1950332	5131.0

```
##
```

```
## Step:  AIC=5071.39
```

```
## attendee_count ~ RSVPyes + RSVPmaybe + RSVPnotGoing + numAllowedOrgs +
##   numApprovedID + event_hour + day_type + event_type + numAllowedOrgs:numApprovedID
```

```
##
```

	Df	Sum of Sq	RSS	AIC
## <none>			1775643	5071.4

```
## - day_type          1      6288 1781931 5071.6
## + orgPlusOnes_num   1      3750 1771893 5072.0
## - RSVPnotGoing      1      8257 1783900 5072.3
## - event_hour        1     15850 1791493 5075.0
## - numAllowedOrgs:numApprovedID 1     56125 1831768 5089.2
## - RSVPyes           1     117891 1893535 5110.2
## - event_type        7     204159 1979802 5126.5
## - RSVPmaybe        1     204958 1980602 5138.8
```

```
FINALmodel <- lm(attendee_count ~ RSVPyes + RSVPmaybe + RSVPnotGoing +
  numAllowedOrgs + numApprovedID + event_hour + day_type +
  event_type + numAllowedOrgs * numApprovedID,
  data = filtered_uva_doorlist)
summary(FINALmodel)
```

```
##
## Call:
## lm(formula = attendee_count ~ RSVPyes + RSVPmaybe + RSVPnotGoing +
##     numAllowedOrgs + numApprovedID + event_hour + day_type +
##     event_type + numAllowedOrgs * numApprovedID, data = filtered_uva_doorlist)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -266.271  -26.114    1.555   26.359  177.114
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -14.03175    18.53470  -0.757  0.44931
## RSVPyes         0.45358     0.07075   6.411 2.87e-10 ***
## RSVPmaybe    -3.81677     0.45154  -8.453 < 2e-16 ***
## RSVPnotGoing   -4.27126     2.51757  -1.697  0.09028 .
## numAllowedOrgs -17.75662     9.56701  -1.856  0.06392 .
## numApprovedID   0.49960     0.03294  15.168 < 2e-16 ***
## event_hour      1.41179     0.60061   2.351  0.01906 *
## day_typedweekend -7.97143     5.38413  -1.481  0.13924
## event_typedfunc 27.39235     9.40590   2.912  0.00372 **
```

```

## event_typeme_plus_3      -39.27527    10.24827   -3.832   0.00014 ***
## event_typemeeting       -18.76676     9.36802   -2.003   0.04558 *
## event_typemixer         -25.98922    10.27991   -2.528   0.01171 *
## event_typeoother         6.83015     10.12899    0.674   0.50036
## event_typerush_event      9.94271    12.80363    0.777   0.43772
## event_typesmall_event    -1.13848     9.64167   -0.118   0.90604
## numAllowedOrgs:numApprovedID -0.07806    0.01765   -4.423  1.15e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 53.56 on 619 degrees of freedom
## Multiple R-squared:  0.6521, Adjusted R-squared:  0.6437
## F-statistic: 77.35 on 15 and 619 DF,  p-value: < 2.2e-16

```